MILITARY SPECIFICATION SHEET

CONNECTOR, FIBER OPTIC, PLUG-RECEPTACLE-ADAPTER STYLE, FIXED SINGLE TERMINUS, THREADED, PLUG, EXPANDED BEAM LENS, EPOXY, FOR 50/125 µm AND 100/140 µm FIBER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the connector described herein shall consist of this specification and the latest issue of MIL-C-83522.

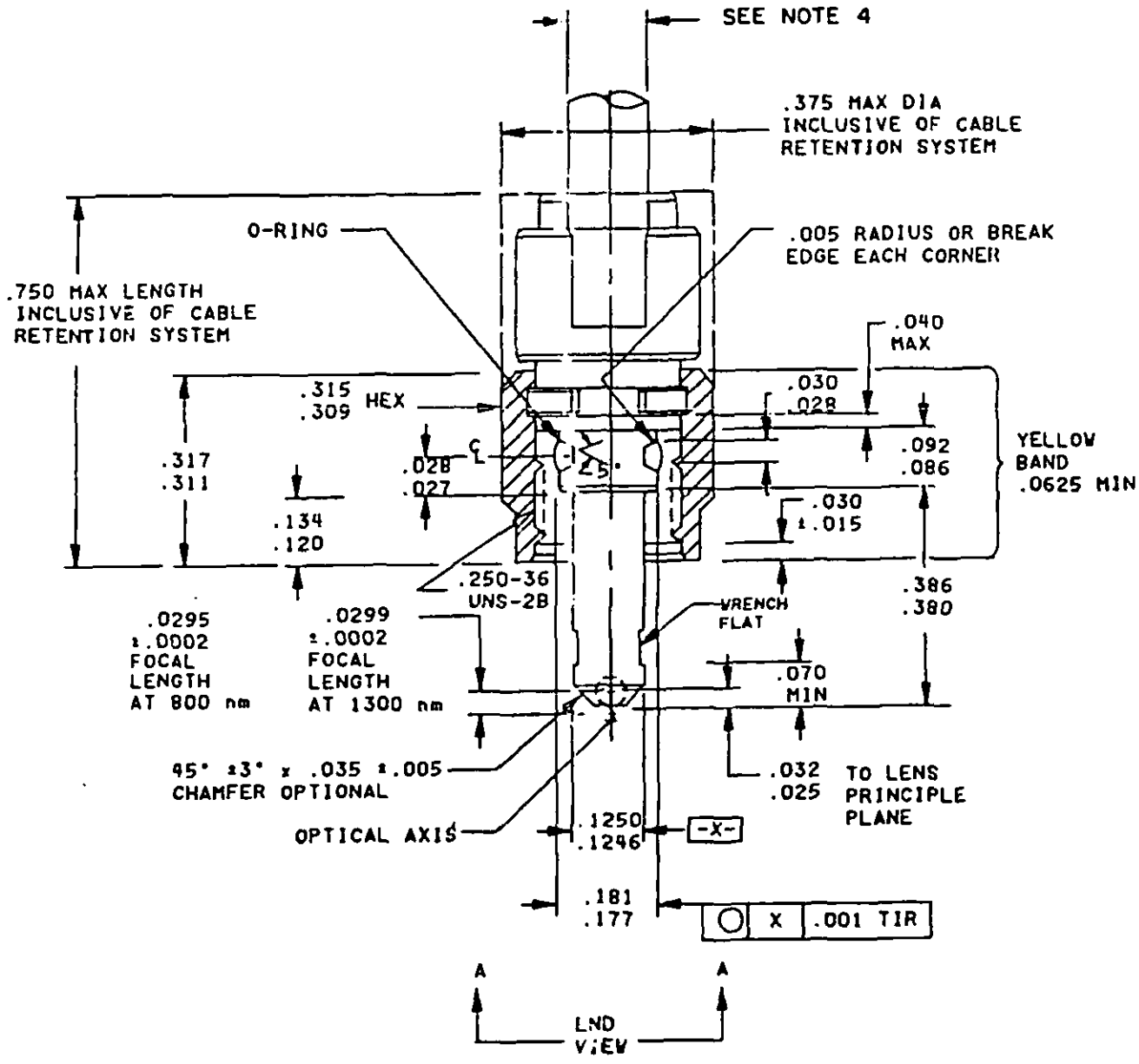
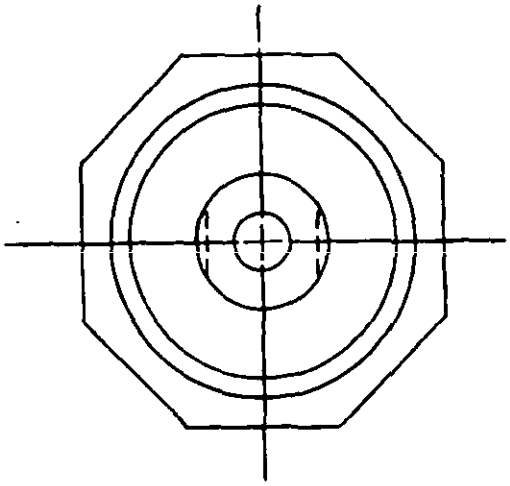


FIGURE 1. Dimensions and configuration.

Ⓑ denotes change

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VIEW A-A

Inches	mm	Inches	mm
.0002	0.005	.092	2.34
.001	0.03	.120	3.05
.005	0.13	.1246	3.165
.015	0.38	.1250	3.180
.025	0.64	.134	3.40
.027	0.69	.177	4.50
.028	0.71	.181	4.60
.0295	0.749	.250	6.35
.0299	0.760	.309	7.85
.030	0.76	.315	8.00
.032	0.81	.317	8.05
.035	0.89	.375	9.53
.040	1.02	.380	9.65
.070	1.78	.386	9.80
.086	2.18	.750	19.05

NOTES:
1. All dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Three holes .016 (0.41 mm) minimum diameter, equally spaced, are required for safety wiring after mating. Location on the coupling nuts optional.
4. Dimensions for fiber optic cable shall be in accordance with DOD-C-85045.

FIGURE 1. Dimensions and configuration - Continued.

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REQUIREMENTS:

Metals: The connector housing and other metal components shall be of nickel silver alloy C79200 in accordance with ASTM-B-151.

Epoxies: Use Trabond 230, Eccobond 1448 or an equivalent epoxy approved by the qualifying activity.

Dimensions and configuration: See figure 1.

Fiber optic cable requirements: In accordance with M85045/2-1C and M85045/2-1D.

Fiber optic contact:

Method of optical alignment: Alignment bushing.

Lens configuration: The lens shall be made from an environmentally resistant material, and its principal plane location and focal length shall be as specified on figure 1.

Coating requirements: The lens shall be anti-reflection coated for operation at the specified wavelength. The residual lens reflection loss per surface after coating shall not exceed 0.1 dB.

Cleaning procedures: If cleaning is necessary, the lens shall be cleanable using lens tissue, a cotton swab, or a camel-hair brush and MIL-C-675 optical antireflective coating cleaning fluid.

Optical requirements:

Number of optical termini: One.

Coupling loss (attenuation): 2 dB maximum per mated pair (steady-state excitation).

Weight: 4 grams maximum.

Polarization: Not applicable.

Safety wire holes: Required.

Force to engage and disengage:

Longitudinal force: 12 pounds maximum.

Ⓑ Torque: 4 inch-pounds maximum.

Ⓑ Coupling proof torque: 5 inch-pounds minimum.

Coupling mechanism retention force: 60 pounds minimum.

Ozone exposure: Not applicable.

Military part number: See table I.

TABLE I. Part numbers.

Part number M83522/6-	Fiber size	Operating wavelength
01	50/125	820
02		1350
03	100/140	820
04		1350

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Plug accessories: Each plug shall be supplied with suitable means for terminating and clamping a single-fiber optical cable of the fiber size and cable diameter range specified. One protective cable-end cap and one set of assembly instructions shall be included. The assembly instructions shall include a listing of the epoxies specified in this specification sheet and a cleaning procedure statement as follows:

> Cleaning procedures: The lens shall be cleaned using lens tissue, a cotton swab, or a camel-hair brush and MIL-C-675 optical antireflective coating cleaning fluid.

NOTE: The plug mates with a MIL-C-83522/4 receptacle; MIL-C-83522/5 receptacle; MIL-C-83522/7 receptacle; MIL-C-83455/8 receptacle; or a MIL-C-83522/3 adapter mated with another MIL-C-83522/6 plug. Connection to any other type plug may result in excessive coupling loss.

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Navy - SH
Air Force - 11, 17, 19, 80, 90, 99
DLA - ES

User activities:
Navy - AS, MC, OS
Air Force - 13, 14

Preparing activity:
Air Force - 85

Agent:
DLA - ES

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